

Linear Algebra-Midterm

Student Name/Number: _____

Question	1	2	3	4	Total
Grade					

1. (25pt) For which values of k and l the following system of linear equations will have no solutions, exactly one solution, and infinitely many solutions?

$$x + y + 5z = l,$$

$$y + (2 + k)z = l,$$

$$x + (3 + k)z = 2l,$$

2. (25pt) Find the inverse, if it exists, for the following matrix:

$$A = \begin{bmatrix} 1 & 0 & -1 & 5 \\ -1 & 1 & 0 & -3 \\ 0 & 2 & -3 & 7 \\ 2 & 1 & -2 & 12 \end{bmatrix}$$

3. (25pt) Use elementary row or column operations to evaluate the determinant of the following matrix:

$$A = \begin{bmatrix} 1 & -1 & 8 & 4 & 2 \\ 2 & 6 & 0 & -4 & 3 \\ 2 & 0 & 2 & 6 & 2 \\ 0 & 2 & 8 & 0 & 0 \\ 0 & 1 & 1 & 2 & 2 \end{bmatrix}$$

4. (25pt) Networks composed of branches and junctions are used as models in many diverse fields such as economics, traffic analysis, and electrical engineering. In such models it is assumed that the total flow into a junction is equal to the total flow out of the junction.

The flow of traffic (in vehicles per hour) through a network of streets is shown in the Figure.

- (a) Solve this system for x_i , $i = 1, 2, 3, 4, 5$.
 (b) Find the traffic flow when $x_3 = 0$ and $x_5 = 1000$.

